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UPGRADED ROS 2 4-DOF INDUSTRIAL ROBOTIC ARM

Xacro-Based Model with Wrist, Gripper and Inertial Properties

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ROS Version:	ROS 2 Humble
Operating System:	Ubuntu 22.04.5 LTS
Gazebo Version:	11.10.2
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Abstract

This project presents an upgraded version of a 4-DOF industrial robotic arm implemented using ROS 2 Humble.

The original 4-DOF robotic arm was first developed using URDF. After successfully validating the basic model, the robot was enhanced using Xacro to provide a more modular and parameterized robot description.

The upgraded model retains the original four revolute joints and four main arm links. A wrist link, gripper base, and two gripper fingers were added to improve the end-effector structure. Visual and collision geometries were defined for the new components.

Approximate mass and inertia properties were also added to the robot links. The upgraded model was validated using `check_urdf`, visualized in RViz2, and its complete TF hierarchy was verified using `tf2_tools`.

The upgraded robot provides a foundation for future work such as Gazebo physics simulation, ROS 2 joint control, inverse kinematics, trajectory planning, and MoveIt 2 integration.

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Chapter 1

Introduction

A robot description is an important component of a ROS 2 robotic system. It defines the links, joints, geometry, and relationships between different components of the robot.

The initial project consisted of a 4-DOF industrial robotic arm implemented using URDF. After completing the basic requirements, the robot was upgraded to provide a more complete industrial robot model.

The upgraded implementation uses Xacro and contains:

- Four primary revolute joints
- Five original structural links
- A wrist link
- A gripper base
- Two gripper fingers
- Visual geometry
- Collision geometry
- Approximate inertial properties
- Extended TF structure

The robot was tested using ROS 2 Humble on Ubuntu 22.04.5 LTS.

Chapter 2

Objectives

The objectives of the upgraded project are:

1. Implement the original 4-DOF industrial robotic arm.
2. Convert the robot description from URDF to Xacro.
3. Introduce parameterized robot dimensions.
4. Add a wrist section to the robot.
5. Add a two-finger gripper.
6. Define visual and collision geometry for the added components.
7. Add approximate mass and inertia properties.
8. Validate the complete Xacro-generated URDF.
9. Visualize the upgraded robot in RViz2.
10. Verify the complete TF tree.
11. Prepare the robot model for future Gazebo simulation.

Chapter 3

Software and Tools

The following software and tools were used during the project.

Table 3.1: Software and Tools

Tool	**Version / Purpose**
Ubuntu	22.04.5 LTS
ROS 2	Humble
URDF	Robot description format
Xacro	Parameterized robot description
RViz2	Robot visualization
robot_state_publisher	TF publishing
joint_state_publisher_gui	Joint state visualization
tf2_tools	TF tree generation
tf2_ros	TF transform verification
check_urdf	URDF validation
Gazebo	11.10.2

Chapter 4

Original Robot Structure

The original 4-DOF robot contained five links:

```
base_link
link1
link2
link3
link4
```

The four primary joints were:

```
joint1
joint2
joint3
joint4
```

The original structure was:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
```

The four primary joints were retained in the upgraded model.

Chapter 5

Conversion from URDF to Xacro

The original URDF was converted into a Xacro-based robot description.

The main Xacro file is:

```
urdf/arm_4dof.xacro
```

Xacro properties were introduced for important robot dimensions.

Examples include:

```
<xacro:property name="base_radius" value="0.30"/>
<xacro:property name="base_height" value="0.10"/>

<xacro:property name="link1_radius" value="0.12"/>
<xacro:property name="link1_length" value="0.60"/>

<xacro:property name="link2_radius" value="0.10"/>
<xacro:property name="link2_length" value="0.50"/>

<xacro:property name="link3_radius" value="0.08"/>
<xacro:property name="link3_length" value="0.40"/>

<xacro:property name="link4_radius" value="0.06"/>
<xacro:property name="link4_length" value="0.30"/>
```

This parameterization makes the robot model easier to modify and maintain.

The Xacro file was converted to URDF using:

```
xacro urdf/arm_4dof.xacro > /tmp/arm_4dof.urdf
```

Chapter 6

Primary Revolute Joints

The four primary joints of the robotic arm are revolute joints.

Table 6.1: Primary Joint Configuration

Joint	**Parent**	**Child**	**Axis**
joint1	base_link	link1	Z
joint2	link1	link2	Y
joint3	link2	link3	Y
joint4	link3	link4	Y

The joint limits are:

Table 6.2: Primary Joint Limits

Joint	**Lower**	**Upper**	**Effort**	**Velocity**
joint1	-3.14	3.14	10	1
joint2	-1.57	1.57	10	1
joint3	-1.57	1.57	10	1
joint4	-3.14	3.14	10	1

Chapter 7

Wrist and End-Effector Upgrade

The original four-link arm was extended by adding a wrist and two-finger gripper.

The upgraded structure is:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
    |
 wrist_link
    |
gripper_base
   / \
  / \
left right
finger finger
```

7.1 Wrist Link

The wrist is represented by:

```
wrist_link
```

It is connected to `link4` using:

```
wrist_joint
```

The wrist joint is a fixed joint.

7.2 Gripper Base

The gripper base is represented by:

```
gripper_base
```

It is connected to the wrist using:

```
gripper_mount_joint
```

7.3 Gripper Fingers

Two fingers were added:

```
left_finger
right_finger
```

The fingers are connected to the gripper base using fixed joints.

The current design provides the structural representation of a two-finger gripper. Active opening and closing control can be implemented as a future enhancement.

Chapter 8

Visual and Collision Geometry

Visual and collision geometry were defined for the complete robot.

The primary arm links use cylindrical geometries.

The wrist uses a cylindrical geometry.

The gripper base uses a box geometry.

The two fingers use box geometries.

The visual geometry is used by RViz2 for displaying the robot, while collision geometry provides physical geometry for future simulation and collision checking.

Chapter 9

Inertial Properties

Approximate inertial properties were added to the robot links.

Each inertial definition contains:

- Mass
- Center of mass
- I_{xx}
- I_{yy}
- I_{zz}
- Cross-inertia terms

Inertial properties were included for:

```
base_link
link1
link2
link3
link4
wrist_link
gripper_base
left_finger
right_finger
```

The values used in this project are approximate model values. They are intended to provide physical parameters for future simulation and are not claimed to represent experimentally measured masses or inertias.

Chapter 10

Xacro Validation

The Xacro model was converted to URDF using:

```
xacro urdf/arm_4dof.xacro > /tmp/arm_4dof.urdf
```

The generated URDF was then validated using:

```
check_urdf /tmp/arm_4dof.urdf
```

The validation result was:

```
robot name is: arm_4dof
---------- Successfully Parsed XML ---------------
root Link: base_link has 1 child(ren)
    child(1): link1
        child(1): link2
            child(1): link3
                child(1): link4
                    child(1): wrist_link
                        child(1): gripper_base
                            child(1): left_finger
                            child(2): right_finger
```

The successful parsing confirms that the upgraded robot description forms a valid connected link tree.

Chapter 11

RViz2 Visualization

The upgraded robot was loaded into RViz2 using `robot_state_publisher`.

The fixed frame was configured as:

```
base_link
```

The RobotModel display used the:

```
robot_description
```

topic.

The four primary joint sliders were used to verify the motion of the arm.

The wrist and gripper remained connected to the arm while the joints were moved.

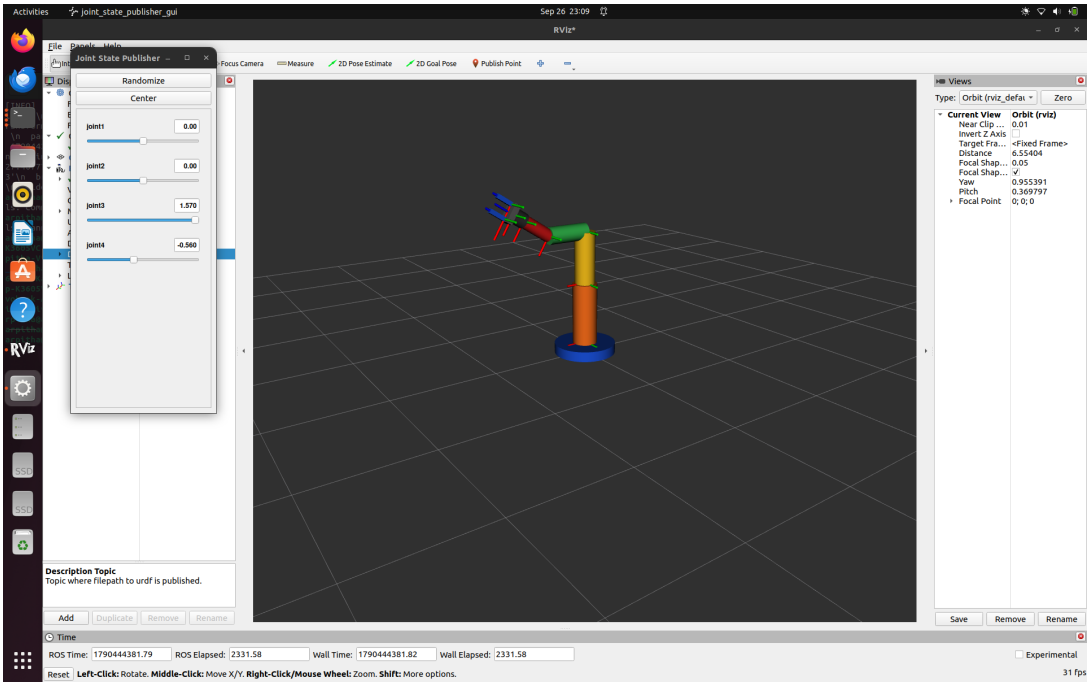


Figure 11.1: Upgraded 4-DOF robotic arm with wrist and two-finger gripper in RViz2

Figure 11.1 shows the upgraded robotic arm visualized in RViz2.

Chapter 12

TF Tree Verification

The TF tree was generated using:

```
ros2 run tf2_tools view_frames
```

The final TF hierarchy is:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
    |
 wrist_link
    |
gripper_base
   / \
  / \
left right
finger finger
```

The TF tree confirms that the original arm and the additional wrist and gripper components are connected.

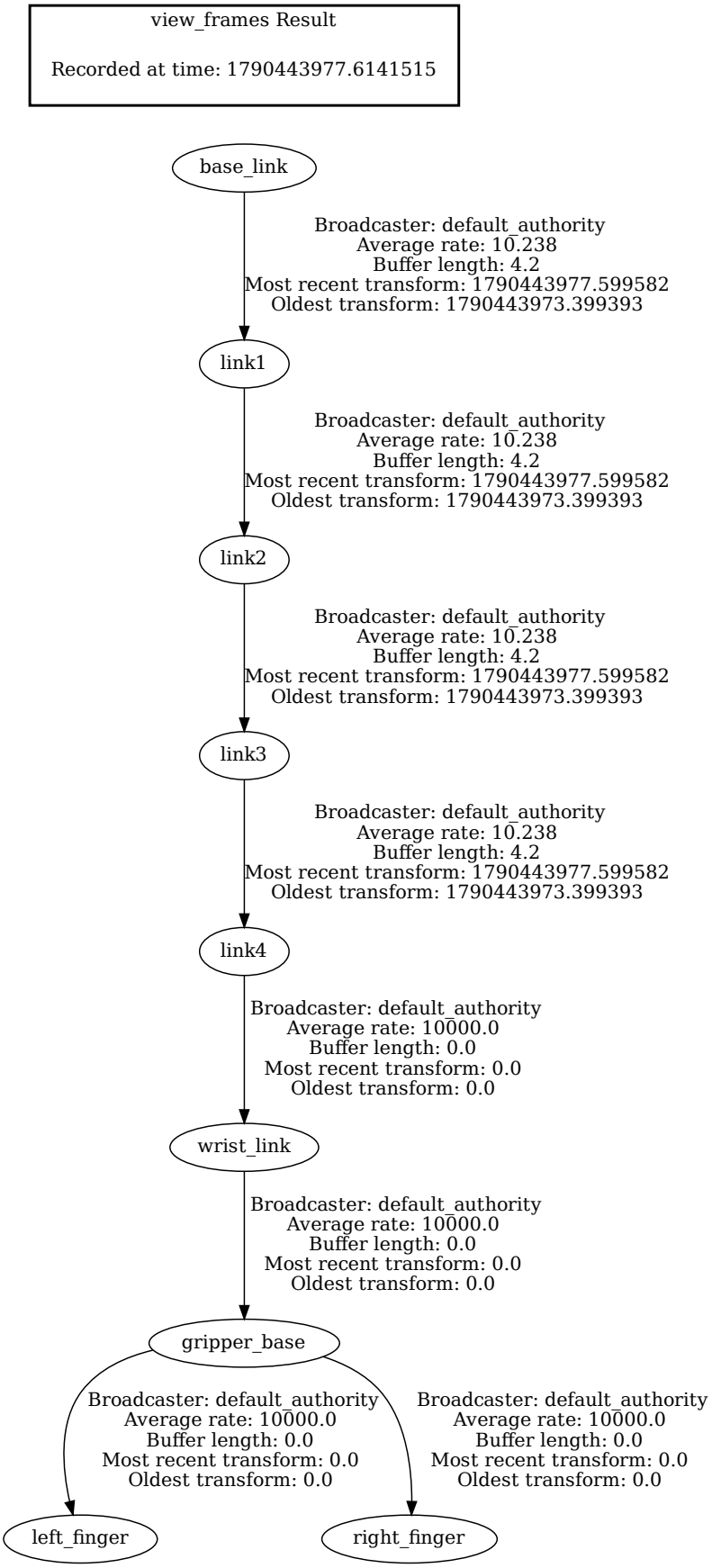


Figure 12.1: TF tree of the upgraded 4-DOF robotic arm

Chapter 13

TF Transform Verification

The connection from the robot base to the right gripper finger was additionally verified using:

```
ros2 run tf2_ros tf2_echo base_link right_finger
```

The complete transform chain is:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
    |
 wrist_link
    |
gripper_base
    |
right_finger
```

The transform query confirms that the right gripper finger can be reached from the robot's base frame through the complete TF chain.

Chapter 14

Gazebo Preparation

Gazebo 11.10.2 is installed in the development environment.

The following ROS packages are available:

```
gazebo_ros
gazebo_ros2_control
gazebo_ros_pkgs
```

A separate launch file was created:

```
launch/gazebo.launch.py
```

The launch file uses the Xacro robot description and prepares the robot for spawning into Gazebo.

The inertial properties added to the links provide physical parameters required for future physics simulation.

At the current stage, Gazebo setup has been prepared, while full physics validation and joint-controller implementation remain future work.

Chapter 15

Comparison with the Basic Model

The main differences between the original assignment model and the upgraded model are shown below.

Table 15.1: Basic and Upgraded Robot Comparison

Basic Model	**Upgraded Model**
URDF	Xacro
5 links	9 links
4 revolute joints	4 primary revolute joints
No wrist	Wrist added
No gripper	Two-finger gripper added
Visual geometry	Visual geometry retained and extended
Collision geometry	Collision geometry retained and extended
No inertial properties	Approximate inertial properties added
Basic TF tree	Extended TF tree
RViz visualization	RViz visualization
TF verification	Extended TF verification
No Gazebo preparation	Gazebo launch preparation

Chapter 16

Project File Structure

The upgraded ROS 2 package is organized as follows:

```
arm_4dof_description/
|
+-- CMakeLists.txt
+-- package.xml
|
+-- urdf/
| +-- arm_4dof.urdf
| +-- arm_4dof_backup.urdf
| +-- arm_4dof.xacro
| +-- arm_4dof_xacro_backup.xacro
| +-- arm_4dof_inertial_backup.xacro
|
+-- launch/
| +-- display.launch.py
| +-- gazebo.launch.py
|
+-- rviz/
```

Chapter 17

Validation Results

Table 17.1: Project Validation

Validation	**Status**
Xacro conversion	Completed
Xacro-generated URDF parsing	Completed
Four primary revolute joints	Completed
Joint limits	Completed
Visual geometry	Completed
Collision geometry	Completed
Wrist addition	Completed
Two-finger gripper	Completed
Inertial properties	Completed
RViz2 visualization	Completed
Joint slider verification	Completed
TF tree generation	Completed
TF connectivity verification	Completed
Gazebo installation verification	Completed
Full Gazebo physics validation	Future work
Joint controller implementation	Future work

Chapter 18

Conclusion

The original 4-DOF industrial robotic arm was successfully implemented using ROS 2 and URDF.

The model was subsequently upgraded using Xacro to make the robot description more parameterized and maintainable.

The upgraded robot retains the original four revolute joints and adds a wrist, gripper base, and two gripper fingers. Visual and collision geometries were defined for the complete model.

Approximate inertial properties were added to prepare the robot for future physics-based simulation.

The upgraded Xacro model was successfully converted to URDF and validated using `check_urdf`. The robot was visualized in RViz2, and the four primary joints were tested using the joint state GUI.

The complete TF tree was generated and verified, including the wrist and two-finger gripper.

The resulting robot model provides a foundation for further development in simulation, control, kinematics, and motion planning.

Chapter 19

Future Scope

The following improvements can be implemented in future work:

1. Complete Gazebo physics simulation.
2. Implement ROS 2 joint controllers.
3. Add active gripper opening and closing.
4. Integrate `ros2_control`.
5. Implement forward kinematics.
6. Implement inverse kinematics.
7. Integrate MoveIt 2.
8. Implement trajectory planning.
9. Add collision-aware motion planning.
10. Simulate object grasping and manipulation.

Appendix A

Important Commands

A.1 Convert Xacro to URDF

```
xacro urdf/arm_4dof.xacro > /tmp/arm_4dof.urdf
```

A.2 Validate Robot

```
check_urdf /tmp/arm_4dof.urdf
```

A.3 Build Package

```
cd ~/ros2_ws
colcon build --packages-select arm_4dof_description
```

A.4 Source Workspace

```
source ~/ros2_ws/install/setup.bash
```

A.5 Launch RViz

```
ros2 launch arm_4dof_description display.launch.py
```

A.6 Generate TF Tree

```
ros2 run tf2_tools view_frames
```

A.7 Verify TF Transform

```
ros2 run tf2_ros tf2_echo base_link right_finger
```

A.8 Launch Gazebo

```
ros2 launch arm_4dof_description gazebo.launch.py
```